

Event-TimeRAF: Event-Aware Retrieval-Augmented Foundation Model for Explainable Air Quality Forecasting Under Concept Drift

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Abstract—Urban air-quality forecasting is important for public-health warnings, environmental management, and city-scale decision support. Time-series foundation models have recently shown promising zero-shot forecasting capability, and TimeRAF demonstrates that retrieval augmentation can supplement a frozen forecasting backbone with useful historical knowledge. However, PM2.5 variation is strongly affected by local context such as meteorology, holidays, high-wind events, heavy rain, wildfire activity, and other environmental disturbances. This paper introduces Event-TimeRAF, an event-aware retrieval-augmented framework for Los Angeles County PM2.5 forecasting. Event-TimeRAF builds a source-preserving knowledge base from official air-quality, weather, calendar, and storm-event records; retrieves causally eligible historical windows; fuses retrieved evidence with tabular forecasters and a frozen Chronos-Bolt foundation model; and generates evidence-grounded explanations under an explicit drift audit. We evaluate the framework on EPA AirData PM2.5, NOAA Global Hourly weather, and hash-verified NOAA Storm Events records for 2019–2024. A lightweight drift-gated Event-TimeRAF variant obtains the strongest numerical result with MSE 26.111, MAE 3.116, RMSE 5.110, and R^2 0.381. The strongest ungated supervised context baseline obtains MSE 26.185, while the full feature-level Event-TimeRAF XGBoost variant obtains MSE 26.712. The frozen Chronos-Bolt baseline obtains MSE 28.941, and retrieval-augmented Chronos reduces it to MSE 28.709. These results show that retrieval is beneficial in selected settings, especially for the frozen foundation-model fusion and drift-gated selection, but the MSE gain over the strongest context baseline is small and not statistically conclusive. Because NOAA Storm Events lacks machine-readable publication timestamps, event-aware results are reported as retrospective event-start sensitivity results rather than strict real-time event forecasts.

Index Terms—Air quality forecasting, PM2.5, retrieval-augmented forecasting, time-series foundation models, TimeRAF, concept drift, explainable forecasting.

I. INTRODUCTION

Fine particulate matter (PM2.5) is a major urban air-quality risk because it is associated with respiratory and cardiovascular health effects [1]. Accurate short-term PM2.5 forecasts can support public-health alerts, exposure reduction, transport planning, and environmental operations. In Los Angeles County, however, PM2.5 dynamics are difficult to model with pollutant history alone. Concentrations can change with wind speed, boundary-layer behavior, rainfall, seasonal activity, holidays, wildfire smoke, and unusual storm or high-wind events. These factors make the forecasting problem both temporal and contextual.

Recent time-series foundation models (TSFMs) aim to improve forecasting generalization by pre-training on large time-

series corpora and transferring the learned forecasting ability to new domains. Representative models include TimesFM, Moirai, MOMENT, Chronos, and Tiny Time Mixers [2]–[6]. In parallel, retrieval-augmented generation has shown that non-parametric memory can help foundation models access relevant external knowledge at inference time [7], [8]. TimeRAF combines these two ideas for zero-shot time-series forecasting by retrieving relevant historical sequences from a curated knowledge base and integrating them into a frozen TSFM through Channel Prompting [9].

Air-quality forecasting raises a domain-specific retrieval question that is not fully addressed by numerical time-series retrieval alone. A historical PM2.5 window may look similar to the current window but occur under different weather or event conditions. Conversely, a window with moderate numerical similarity may be more informative if it shares high-wind, rainfall, holiday, smoke, or stagnation context. This motivates a retrieval framework that treats source records, weather, calendar information, and distribution-shift indicators as part of the forecasting evidence rather than as unrelated metadata.

This paper introduces Event-TimeRAF, an event-aware retrieval-augmented framework for explainable PM2.5 forecasting under concept drift. The design is inspired by TimeRAF, but it is adapted to an official-source environmental forecasting setting. The implemented system uses leakage-safe retrieval over historical PM2.5 windows, weather and calendar context, source-recorded NOAA Storm Events, transparent drift indicators, direct XGBoost forecasters, a frozen Chronos-Bolt baseline, and evidence-grounded explanations. The framework answers three practical questions: (1) how can official environmental records be converted into a retrieval-compatible knowledge base; (2) how should retrieval be constrained so that future target information is not leaked; and (3) when does retrieval help beyond strong weather-calendar supervised baselines?

The main contributions are as follows:

- We formulate an event-aware retrieval-augmented PM2.5 forecasting framework that combines time-series windows, weather summaries, calendar context, storm-event records, and drift indicators.
- We implement a reproducible official-source pipeline using EPA AirData PM2.5, NOAA Global Hourly weather, and hash-verified NOAA Storm Events records for Los Angeles County during 2019–2024.
- We define leakage-safe retrieval eligibility rules that prevent candidate windows from overlapping the query

context or future target.

- We evaluate random retrieval, cosine retrieval, event-aware hybrid retrieval, drift-aware feature fusion, frozen Chronos-Bolt, and retrieval-augmented Chronos-Bolt under a controlled evaluation protocol.
- We report a conservative empirical finding: retrieval improves the frozen Chronos-Bolt MSE slightly, but the best overall model is a strong XGBoost weather-calendar context baseline.

The findings therefore do not indicate uniform gains from retrieval. Instead, the evaluation identifies the settings in which event-aware retrieval is useful, the settings in which it is not, and the additional evidence required before the method can support operational event-aware air-quality forecasting.

II. LITERATURE REVIEW AND RELATED WORK

A. Time-Series Forecasting

Time-series forecasting has traditionally been studied through statistical models, machine-learning regressors, and deep neural networks. Autoregressive and seasonal models remain useful baselines because they are interpretable and efficient. Gradient-boosted tree models such as XGBoost and LightGBM are widely used for tabular forecasting with lagged and rolling features [10], [11]. Recurrent models such as LSTM and GRU can capture sequential dependence [12], [13]. Transformer-based models then expanded the long-sequence forecasting literature. Informer reduces attention cost for long inputs [14], Autoformer and FEDformer combine decomposition with long-range dependency modeling [15], [16], Crossformer and iTransformer focus on multivariate channel structure [17], [18], and TimesNet models temporal variation through 2D transformations [19]. At the same time, DLinear showed that simple linear decomposition models can outperform more complex Transformer variants on some long-term forecasting benchmarks [20]. PatchTST later used patching and channel-independent designs to improve long-lookback efficiency [21]. For air-quality forecasting, these approaches are helpful, but they may still be sensitive to distribution shifts and unobserved external events.

B. Time-Series Foundation Models

TSFMs aim to provide transferable forecasting ability across datasets and domains. TimesFM uses a decoder-only architecture for time-series forecasting [2]. Moirai proposes a universal forecasting Transformer trained on a large-scale open time-series archive [3], [22]. MOMENT provides a family of open time-series foundation models for general-purpose time-series tasks [4]. Chronos converts time-series values into tokens and trains language-model architectures for probabilistic forecasting [5]. TTM proposes compact pre-trained models for efficient zero/few-shot forecasting [6]. Timer and Timer-XL study generative pre-training and long-context Transformer scaling for unified time-series forecasting [23], [24]. UniTime and Time-LLM connect language-model ideas with cross-domain or reprogrammed time-series forecasting [25], [26]. These models motivate the use of pre-trained forecasting backbones, but practical domain adaptation is still difficult

when the downstream series responds strongly to local events and environmental conditions.

C. Retrieval-Augmented Forecasting

Retrieval-augmented generation first became popular in NLP by combining parametric models with non-parametric memory [7]. Dense Passage Retrieval (DPR) introduced dense dual-encoder retrieval for open-domain question answering [8]. Retrieval ideas have also been adapted to time-series settings through retrieval-augmented diffusion forecasting and RAG-assisted time-series generation workflows [27], [28]. TimeRAF adapts the retrieval-augmentation idea to zero-shot time-series forecasting [9]. The TimeRAF study frames retrieval-augmented forecasting around three design questions: what knowledge base should support the forecaster, how relevant time-series knowledge should be retrieved, and how retrieved candidates should be integrated into a frozen TSFM. Its solution constructs a time-series knowledge base, retrieves top- k candidate windows using a learnable MLP-based dual-encoder retriever, and combines retrieved candidates through Channel Prompting. TimeRAF reports TTM-Base as its main backbone, a context length of 512, a forecast horizon of 96 in most experiments, and $k = 8$ as the default number of retrieved candidates. Its ablations compare random retrieval, cosine retrieval, learnable retrieval, alternative integration mechanisms, candidate counts, knowledge-base choices, and backbone choices.

Event-TimeRAF follows the TimeRAF principle but changes both the knowledge source and the target domain. Instead of relying only on numerical time-series retrieval, it uses a hybrid knowledge base containing historical PM2.5 windows, weather summaries, calendar context, and event evidence. The implemented system is TimeRAF-inspired rather than a full reproduction: it evaluates random retrieval, cosine retrieval, event-aware hybrid retrieval, drift-aware feature fusion, frozen Chronos-Bolt, and retrieval-augmented Chronos-Bolt forecast fusion, while a learnable dual-encoder retriever and full Channel-Prompting neural integration remain future extensions.

D. Air-Quality Forecasting and Data Sources

PM2.5 forecasting often uses historical pollutant measurements together with meteorological factors such as temperature, humidity, wind speed, precipitation, and pressure. Recent PM2.5 studies and reviews show a shift from single-station recurrent models toward deep learning, attention mechanisms, and hybrid spatiotemporal workflows [29], [30]. These studies motivate the use of meteorology, temporal context, and spatial/event structure, but they usually evaluate learned forecasters rather than source-audited retrieval evidence. The US Environmental Protection Agency publishes quality-controlled hourly observations through AirData and the Air Quality System [31]. NOAA's Integrated Surface Database provides hourly meteorological observations from surface stations [32]. NOAA also publishes structured Storm Events records and historical Hazard Mapping System fire and smoke products [33], [34]. These official sources support a source-preserving

Los Angeles County forecasting pipeline without requiring manually authored event records.

E. Explainable and Event-Aware Forecasting

Explainability is important in environmental forecasting because users need to understand why a model predicts a rise or fall in pollution. SHAP values provide a model-agnostic framework for estimating feature contributions [35]. However, explanations based only on feature importance may miss temporal and contextual cues. Event-TimeRAF therefore generates explanations from structured evidence: recent PM2.5 persistence, weather flags, retrieved similar historical windows, retrieved event records, calendar context, and drift indicators. The explanation module uses templates grounded in stored evidence, not unrestricted language-model generation.

III. CRITICAL GAPS AND CORE CONTRIBUTIONS

A. Critical Gaps in Prior Studies

Existing forecasting approaches leave several gaps for event-sensitive urban air quality forecasting.

Gap 1: numerical history alone is not enough. Statistical, deep, and foundation forecasting models can learn temporal regularities, but PM2.5 changes often depend on weather, calendar behavior, and source-recorded environmental events. A model that ignores those factors can retrieve or learn patterns that are numerically similar but contextually different.

Gap 2: retrieval must be leakage-safe. Time-series retrieval can easily introduce look-ahead bias if retrieved candidates overlap the query input or target period. Environmental forecasting studies also risk using post-hoc event labels as if they were available before the forecast origin.

Gap 3: event availability is rarely audited. Public event databases are useful but not always operationally timestamped. NOAA Storm Events is official, but it does not provide a machine-readable publication timestamp. Without an explicit availability rule, event-aware forecasting claims may overstate real-time usability.

Gap 4: foundation-model gains need domain verification. TSFMs can be strong zero-shot forecasters, but their advantage is not guaranteed in a local PM2.5 task where supervised weather-calendar features are highly informative. Retrieval-augmented foundation-model claims therefore require a frozen-backbone baseline and a retrieval-augmented variant evaluated on the same test origins.

Gap 5: explanations should be evidence-grounded. Feature-importance scores alone do not explain whether a forecast is linked to weather shift, retrieved historical cases, event context, or drift. A deployable environmental forecasting workflow needs traceable evidence rather than unrestricted generated text.

B. Core Contributions

The proposed Event-TimeRAF framework addresses these gaps through the following components:

- *Context-aware retrieval knowledge base:* a source-preserving knowledge base that combines PM2.5 windows, weather summaries, calendar signatures, and NOAA Storm Events context.
- *Temporally leakage-safe retrieval:* a strict candidate-filtering rule that allows a candidate only when its future target window finishes before the query lookback begins.
- *Audited event availability:* a data-readiness audit that records event source, delivery mode, source overlap, category coverage, and whether the event timestamps are strict publication times or retrospective event-start assumptions.
- *Dual forecasting evaluation:* direct XGBoost feature-level fusion is evaluated alongside frozen Chronos-Bolt and validation-weighted retrieval-augmented Chronos fusion.
- *Evidence-based explanation layer:* forecast explanations are constructed from stored retrieval, weather, event, drift, and local model contribution evidence.

Unlike a full TimeRAF reproduction, this work does not train a dual-encoder retriever or implement internal Channel Prompting. Instead, it provides an audited domain adaptation of the retrieval-augmentation idea and tests whether the idea is useful under official-source PM2.5 forecasting constraints.

IV. PROBLEM FORMULATION

Let x_t denote the PM2.5 measurement at time t . For a forecast time t , the historical air-quality lookback window is

$$X_t = [x_{t-L+1}, x_{t-L+2}, \dots, x_t], \quad (1)$$

where L is the lookback length. Let W_t denote weather covariates aligned with the lookback window, C_t denote calendar features, and E_t denote event context obtained from the NOAA Storm Events knowledge base. The target is the future PM2.5 sequence

$$Y_{t+1:t+H} = [x_{t+1}, x_{t+2}, \dots, x_{t+H}], \quad (2)$$

where H is the forecasting horizon.

The forecasting goal is to learn a function

$$\hat{Y}_{t+1:t+H} = F_\theta(X_t, W_t, C_t, E_t, R_t^{ts}, R_t^{ev}, D_t), \quad (3)$$

where R_t^{ts} is retrieved time-series knowledge, R_t^{ev} is retrieved event knowledge, and D_t contains concept-drift indicators. The implemented target setting is $L = 168$ hours and $H = 24$ hours.

The evaluation uses a time-based split:

$$\mathcal{T}_{train} < \mathcal{T}_{val} < \mathcal{T}_{test}, \quad (4)$$

where the earliest 70% of observations are used for training, the next 15% for validation, and the latest 15% for testing. Random splitting is not used because it would leak future temporal information into training.

Table I summarizes the essential notation.

TABLE I
MAIN SYMBOLS USED IN THE PROPOSED FORMULATION

Symbol	Meaning
L	Lookback window length
H	Forecast horizon
X_t	Historical PM2.5 sequence at time t
W_t	Weather covariates
C_t	Calendar/context features
E_t	Event context available at the forecast origin
R_t^{ts}	Retrieved time-series knowledge
R_t^{ev}	Retrieved event knowledge
D_t	Drift indicators
$\hat{Y}_{t+1:t+H}$	Predicted future PM2.5 values

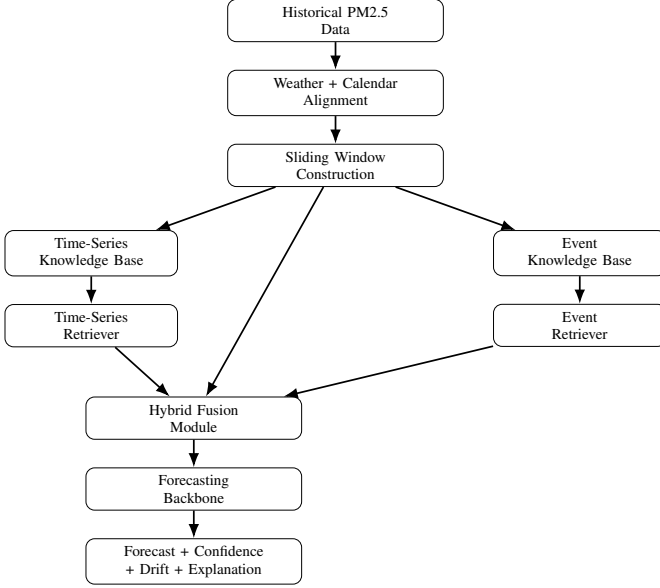


Fig. 1. Event-TimeRAF forecasting architecture.

V. METHODOLOGY

A. Overview

Fig. 1 shows the Event-TimeRAF pipeline. The first step is to align air-quality observations with weather, calendar, and availability-filtered event features. The system then creates sliding windows and a time-series knowledge base. The implemented hybrid retriever ranks historical windows using PM2.5, weather, calendar, and event-context similarity. A separate source-record lookup links recent event identifiers to explanations; it does not supply future event labels to the model. The retrieved evidence is combined with structured features to generate a forecast, uncertainty proxy, drift score, and explanation.

For each forecast origin t , Event-TimeRAF maps the aligned context into a forecast through

$$\hat{Y}_{t+1:t+H} = F_{\theta}(\phi(X_t, W_t, C_t, E_t), G_t, D_t), \quad (5)$$

where $\phi(\cdot)$ denotes the tabular feature construction, G_t denotes retrieved historical evidence, and D_t denotes drift evidence. The key methodological constraint is that every element of G_t and D_t must be available no later than the forecast origin.

B. Relationship to TimeRAF

TimeRAF retrieves top- k candidate time-series windows from a curated knowledge base and incorporates the retrieved candidates using Channel Prompting [9]. It uses a learnable MLP-based dual-encoder retriever, dot-product similarity, a frozen TSFM backbone, and candidate augmentation during retriever training. Event-TimeRAF keeps the retrieval-augmentation principle, but changes the application setting from general zero-shot benchmark forecasting to official-source Los Angeles County PM2.5 forecasting with weather, calendar, event, drift, and explanation constraints.

Table II gives the direct methodological comparison. The comparison is not a head-to-head performance ranking because the two studies use different datasets, forecast horizons, and evaluation protocols. The key distinction is that TimeRAF is a full neural retrieval-augmented TSFM architecture, whereas Event-TimeRAF is a domain adaptation that prioritizes official data provenance, temporally leakage-safe retrieval, event-availability auditing, and evidence-grounded environmental explanations. A dual-encoder retriever, candidate augmentation, and full internal Channel Prompting are therefore treated as future extensions rather than completed components in this implementation.

C. Dataset Construction

Hourly PM2.5 observations are taken from US EPA Air-Data/AQS parameter 88101 for Los Angeles County [31]. The data audit selects one monitoring site using coverage and continuity criteria when a single site satisfies the coverage criterion. Because no individual monitor met this criterion, the target is the hourly county median across official Los Angeles County AQS monitors. Timestamps are converted to *America/Los_Angeles* with explicit daylight-saving handling. Short input gaps may be filled only from past-available observations; targets are never interpolated. Weather covariates come from the nearest suitable NOAA Integrated Surface Database station [32]. Calendar features include hour, weekday, weekend, month, season, US federal holidays, and California holidays.

All normalization statistics are fitted on the training split only. The same normalization procedure is used for query windows and retrieved candidate windows, following the TimeRAF requirement that input and retrieved sequences use consistent preprocessing.

D. Time-Series Knowledge Base

The time-series knowledge base contains historical PM2.5 windows. Each record stores a window identifier, start time, end time, input window values, future window values, weather summary, calendar summary, normalized window vector, data split, and normalization identifier. Knowledge-base origins are separated by 192 hours, equal to the 168-hour input and 24-hour future combined. For every query, a candidate is eligible only when its complete future ends before the query input begins. Validation and test retrieval is additionally restricted to training-history windows. These rules exclude self-matches, near-duplicate windows, and overlap with the query context.

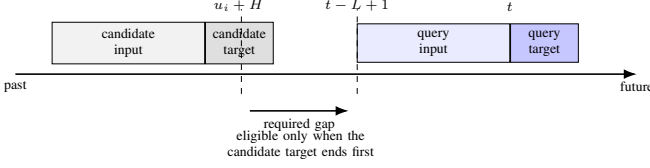


Fig. 2. Leakage-safe retrieval timeline used to enforce temporal eligibility.

Formally, the knowledge base is

$$\mathcal{K} = \{(u_i, X_i, Y_i, m_i, c_i, e_i)\}_{i=1}^N, \quad (6)$$

where u_i is the candidate origin, X_i and Y_i are the candidate lookback and future windows, m_i is a weather summary, c_i is a calendar summary, and e_i is the event-context summary available at u_i . For a query origin t , the temporally eligible set is

$$\mathcal{A}_t = \{i : u_i + H < t - L + 1, i \in \mathcal{T}_{train}\}, \quad (7)$$

for validation and test queries. The strict inequality in (7) ensures that a candidate's target period finishes before the query input period begins.

Given a query window X_t , the time-series retriever returns

$$R_t^{ts} = \{r_{t,1}^{ts}, r_{t,2}^{ts}, \dots, r_{t,k}^{ts}\}, \quad (8)$$

where k is the number of retrieved candidates. The default setting is $k = 8$, with sensitivity analysis for $k \in \{1, 4, 8, 16\}$.

E. Event Knowledge Base

The event knowledge base contains source-preserving records relevant to Los Angeles County. NOAA Storm Events is the required structured source, while NOAA Hazard Mapping System fire and smoke records remain an optional enrichment [33], [34]. Stored fields include event identifier, event start and end, information-availability assumption, location, category, source, source URL, title, and summary. Event categories include wildfire, smoke, excessive heat, high wind, heavy rain, flood, dust, and other weather-related records.

Model inputs represent events with hourly publication-assumption counts, rolling 24-hour, 72-hour, and 168-hour counts, category-specific 72-hour counts, active-event counts, and an event-burst ratio. Recent source event identifiers are retained only as explanation evidence. Every event used as a model input must be available by the forecast origin under a recorded availability rule. Evaluation distinguishes events published in the prior 72 hours, events active at the origin, and recorded event intervals overlapping the target horizon. Target overlap is used only as a post-hoc label and never as a model input. NOAA Storm Events does not provide a machine-readable publication timestamp; consequently, the implementation records event start as a retrospective availability assumption. Results that use this assumption are labeled sensitivity experiments rather than strict operational forecasts. Strict experiments require source records with genuine publication or issuance timestamps.

F. Hybrid Retrieval Score

The hybrid retrieval score combines time-series similarity, weather relevance, calendar relevance, and event relevance:

$$s(q, i) = \alpha s_{ts}(q, i) + \beta s_w(q, i) + \gamma s_c(q, i) + \delta s_e(q, i), \quad (9)$$

where s_{ts} is time-series similarity, s_w is weather similarity, s_c is calendar similarity, and s_e is event relevance. The implemented weights are $\alpha = 0.5$, $\beta = 0.2$, $\gamma = 0.1$, and $\delta = 0.2$.

The retrieved set is selected as

$$G_t = \text{TopK}_{i \in \mathcal{A}_t} s(t, i). \quad (10)$$

For retrieval-only baselines, the forecast is the uniform or similarity-weighted average of the retrieved candidates' future PM2.5 windows. For feature-level models, the retrieved futures and candidate metadata are summarized as mean, spread, similarity, and event-match features.

G. Fusion and Forecasting Backbone

The feature-level fusion method combines PM2.5 lag/rolling features, weather features, calendar features, retrieved time-series summary features, event features, and drift indicators. The tabular forecasting backbone uses 24 direct XGBoost regressors, one for each forecast horizon, providing an efficient and competitive supervised baseline. Frozen Chronos-Bolt is evaluated separately, both without retrieval and with validation-weighted forecast-level fusion between Chronos and retrieved historical forecasts. Unlike TimeRAF, the evaluated architecture does not modify the foundation model's internal representation through Channel Prompting.

The tabular fusion stage includes a lightweight drift-conditioned selector. It compares the strongest validation candidates separately on drift and non-drift validation origins, then applies the selected source to the corresponding test origins:

$$\hat{Y}_t^G = \begin{cases} \hat{Y}_t^a, & D_t = 0, \\ \hat{Y}_t^b, & D_t = 1, \end{cases} \quad (11)$$

where a and b are chosen only from validation errors. This gate does not train a new model; it tests whether a transparent drift-specific selection rule can avoid applying the same forecaster in all regimes.

The Chronos-Bolt retrieval variant uses forecast-level fusion rather than internal Channel Prompting. Let $\hat{Y}_t^C$ be the frozen Chronos forecast and $\hat{Y}_t^R$ be the retrieved historical forecast. The fused prediction is

$$\hat{Y}_t^{CR}(\omega) = \omega \hat{Y}_t^C + (1 - \omega) \hat{Y}_t^R, \quad (12)$$

where $\omega \in \{0, 0.25, 0.5, 0.75, 1.0\}$ is selected on the validation split. Validation selected $\omega = 0.75$.

H. Concept Drift Detection

The implementation represents operational distribution shift using transparent indicators rather than claiming perfect concept-drift identification. The indicators are recent mean

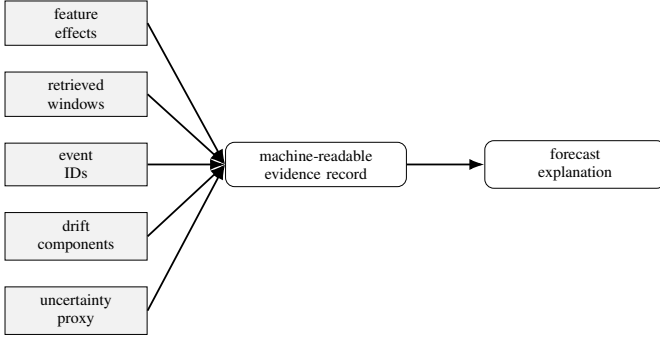


Fig. 3. Evidence-grounded explanation flow used by Event-TimeRAF.

shift, recent variance shift, retrieval similarity drop, weather-pattern shift, and event-burst ratio. A simple drift score can be defined as

$$D_t = \lambda_1 z_{\mu,t} + \lambda_2 z_{\sigma,t} + \lambda_3 (1 - \bar{s}_{ts,t}) + \lambda_4 z_{w,t} + \lambda_5 b_{event,t}, \quad (13)$$

where $z_{\mu,t}$ and $z_{\sigma,t}$ are normalized mean and variance shifts, $\bar{s}_{ts,t}$ is average retrieval similarity, $z_{w,t}$ is weather-pattern shift, and $b_{event,t}$ is an event-burst score. The implemented components are robustly scaled against training data and averaged with equal weight; the alert threshold is selected from validation scores.

I. Explanation Generation

The explanation module generates evidence-based explanations. It uses signed local XGBoost contributions averaged across the 24 direct horizon models, weather flags, retrieved historical windows, retrieved event records, calendar context, forecast direction, component-level drift evidence, and an uncertainty proxy combining retrieved-trajectory spread with validation residual error. It does not generate explanations from an unrestricted language model. Each explanation is traceable to stored machine-readable evidence.

Table II also clarifies which parts of TimeRAF were implemented directly and which parts were adapted for the environmental setting.

VI. EXPERIMENTAL SETUP

This section describes the reported experiment. All model-selection choices were fixed using the training and validation partitions before the test results were examined. Artifact consistency was verified using run identifier 20260723T112033170131Z.

A. Data Sources and Audit

The study period is 2019–2024. PM2.5 observations were taken from US EPA AirData/AQS parameter 88101 for Los Angeles County [31]. No single monitoring site was used as the target series; the audited target is the hourly county median across official Los Angeles County AQS monitors. This fall-back selected 06–037–COUNTY, with 52,602 observed target hours and observed PM2.5 coverage of 0.9997. The PM2.5 unit was consistently recorded as micrograms per cubic meter.

Weather covariates were taken from NOAA Global Hourly/ISD through the official NOAA Open Data Dissemination public buckets [32]. The selected station was 72297023129, Long Beach / Daugherty Field / Airport, 10.28 km from the PM2.5 target centroid. The complete weather-feature coverage after allowed past-only short-gap filling was 0.9942, and the raw measured core coverage was 0.9905.

Environmental events were loaded from hash-verified NOAA Storm Events annual detail archives [33]. Because direct access to NCEI was unavailable in the execution environment, the experiment used a locally staged cache containing unchanged NOAA annual archives and a source manifest with official URLs, byte counts, and SHA-256 hashes. The audit retained the delivery mode `verified_official_noaa_cache`. The event audit found 117 event days, 2,192 source-overlap days, and three qualifying event categories. NOAA Storm Events does not expose machine-readable publication timestamps; therefore, event-aware outputs use the recorded event start time as a retrospective availability assumption. They are not strict real-time event forecasts.

B. Data and Code Availability

The raw public records are available from EPA AirData, NOAA Global Hourly/ISD, and NOAA Storm Events [31]–[33]. The experiment used a locally staged NOAA Storm Events cache because direct NCEI access was unavailable in the execution environment; every cached NOAA file was verified against the generated source manifest. Source code is available at <https://github.com/shasabbir/event-time-raf>.

C. Forecasting Task and Split

The forecasting task predicts PM2.5 from one to 24 hours ahead using a 168-hour lookback window. The split is chronological: the earliest 70% of valid windows are used for training, the next 15% for validation, and the latest 15% for testing. The final test set contains 7,199 forecast origins and 172,776 forecasted hourly target points. Random splitting is not used.

D. Models

The evaluated models are persistence, daily seasonal naive, weekly seasonal naive, a direct XGBoost model using PM2.5 features, a direct XGBoost context model using PM2.5, weather, and calendar features, random retrieval, cosine retrieval, XGBoost with cosine retrieval features, Event-TimeRAF without drift features, full Event-TimeRAF with event and drift features, frozen Chronos-Bolt, and retrieval-augmented Chronos-Bolt forecast fusion. The foundation-model evaluation uses the `amazon/chronos-bolt-small` checkpoint.

E. Implementation and Hyperparameters

The experiment was executed in a cloud-hosted Linux 6.12.90 environment using Python 3.12.13. The runtime was

TABLE II
TIMERAF VERSUS EVENT-TIMERAF

Aspect	TimeRAF	Event-TimeRAF
Primary objective	Improve zero-shot time-series forecasting by augmenting a frozen TSFM with retrieved external time-series knowledge.	Test whether the TimeRAF retrieval idea helps an official-source, event-sensitive PM2.5 forecasting task under drift and availability constraints.
Forecasting domain	Multi-domain time-series benchmarks, with zero-shot generalization as the main setting.	Los Angeles County hourly PM2.5 forecasting for 2019–2024.
Knowledge base	Curated time-series windows built from multi-domain or domain-specific datasets, normalized consistently with the query.	Hybrid official-source knowledge base containing PM2.5 windows, NOAA weather summaries, calendar signatures, and NOAA Storm Events context.
Retriever	Learnable MLP-based dual encoder with dot-product scores and forecaster-guided retriever learning.	Random, cosine, and transparent hybrid similarity retrievers; no learned dual encoder is used.
Candidate count	Uses top- k retrieval; the TimeRAF study uses $k = 8$ as the default and studies candidate-count sensitivity.	Uses $k = 8$ as the default and reports sensitivity for $k \in \{1, 4, 8, 16\}$.
Knowledge integration	Channel Prompting integrates retrieved candidates into the frozen TSFM representation.	Feature-level fusion for XGBoost models, validation-selected drift gating, and validation-weighted forecast-level fusion for frozen Chronos-Bolt.
Backbone	Frozen TSFM backbone, with TTM-Base reported as the principal backbone.	Direct XGBoost horizon models and frozen amazon/chronos-bolt-small.
Leakage control	Uses non-overlapping knowledge-base windows and retrieval constraints, including avoiding same-dataset retrieval during training.	Requires each candidate’s target window to end before the query lookback begins; validation and test retrieval are restricted to training-history windows.
Event handling	Not an explicit component of the base method.	Adds availability-tagged storm-event features and source-record evidence; event results are labeled retrospective when strict publication timestamps are absent.
Drift handling	Not a central module in the base method.	Adds explicit mean, variance, retrieval-similarity, weather, and event-burst drift indicators.
Explainability	Provides retrieved-case visualizations to inspect how retrieved knowledge supports forecasting.	Stores evidence-grounded explanations using feature effects, retrieved windows, event IDs, drift components, and uncertainty proxies.
Implementation status	Full proposed neural architecture with learnable retrieval and Channel Prompting.	TimeRAF-inspired domain implementation; learnable retriever and internal Channel Prompting remain future work.

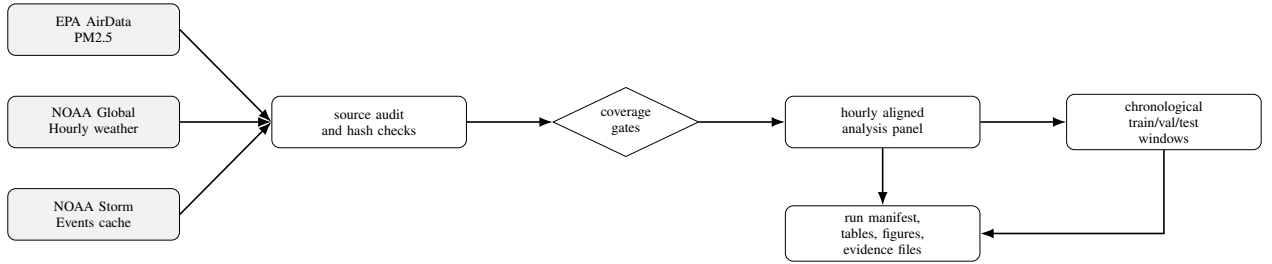


Fig. 4. Official-source data audit and experimental artifact flow.

TABLE III
DATA-READINESS AUDIT

Gate	Value	Status
PM2.5 observed coverage	0.9997	Pass
Weather complete coverage	0.9942	Pass
Event days	117	Pass
Event overlap days	2,192	Pass
Qualifying event categories	3	Pass
Strict event availability	False	Caveat

1,576.87 seconds. Key package versions recorded in the run manifest include NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, XGBoost 3.2.0, PyTorch 2.10.0+cu128, and chronos-forecasting 2.3.1. Table VI reports the principal hyperparameters.

F. Metrics and Statistical Checks

The primary metrics are MSE, MAE, RMSE, MAPE, sMAPE, and R^2 . MSE is retained for comparability with the TimeRAF evaluation, while MAE and RMSE provide more interpretable PM2.5 error scales. Subset metrics are reported only when at least 50 forecast origins are available. The event subset contains 532 origins, the active-event subset contains 293 origins, and the drift subset contains 560 origins, so all pre-specified subsets satisfy the minimum sample-size criterion. Pairwise comparisons use paired block bootstrap intervals with 24-hour blocks and 500 resamples.

G. Position Relative to Existing Studies

Direct numerical comparison with existing TSFM and air-quality papers is not valid because they use different datasets, forecast horizons, variables, and evaluation protocols. Table VII therefore compares methodological coverage rather than claiming cross-paper performance dominance.

TABLE IV
DATASET AND SOURCE SUMMARY

Data group	Source	Evaluated scope	Use in pipeline
PM2.5	EPA AirData/AQS hourly parameter 88101	Los Angeles County, 2019–2024; 52,602 observed target hours	Target sequence and lag/rolling features
Weather	NOAA Global Hourly/ISD, station 72297023129	2019–2024; 0.9942 usable core coverage	Temperature, humidity, wind, pressure, and rain features
Events	NOAA Storm Events annual detail archives	204 Los Angeles records; 117 event days	Retrospective event-start features and explanation evidence
Calendar	Generated from timestamps and public holiday rules	2019–2024 hourly calendar	Hour, weekday, weekend, month, season, and holiday features

TABLE V
EVALUATED MODEL FAMILIES

ID	Description
M00–M02	Persistence and seasonal naive baselines
M03–M04	XGBoost PM2.5-only and context models
M05–M06	Random and cosine retrieval-only baselines
M07	XGBoost with cosine retrieval features
M08–M09	Event-TimeRAF without/with drift indicators
M10–M11	Frozen Chronos-Bolt and retrieval fusion

VII. RESULTS AND DISCUSSION

Table VIII reports the test-set results. The strongest numerical model is the drift-gated Event-TimeRAF variant M12, which obtains MSE 26.111, MAE 3.116, RMSE 5.110, and R^2 0.381. M12 is a prediction-level selector over already trained forecasters: outside drift it follows the validation-selected non-drift source, and during drift it follows the validation-selected drift source. The strongest ungated supervised context model is M04, which uses PM2.5 history, weather, and calendar features and obtains MSE 26.185 and MAE 3.125. The full feature-level Event-TimeRAF XGBoost variant M09 obtains MSE 26.712 and MAE 3.149. Therefore, the results support a modest drift-gated improvement, but it does not support a claim that the full feature-level Event-TimeRAF model uniformly dominates the strongest context baseline.

The frozen Chronos-Bolt evaluation completed successfully. Validation selected a retrieval-fusion weight of 0.75. On the test set, retrieval-augmented Chronos reduces MSE from 28.941 to 28.709, an improvement of about 0.8% relative to frozen Chronos-Bolt. The paired block-bootstrap comparison gives an MSE difference of -0.233 with a 95% interval from -0.665 to 0.152, so the direction is favorable but not conclusive under this uncertainty estimate. MAE changes from 3.205 to 3.209, which is practically flat.

Table IX summarizes the main ablation comparisons. The clearest supported gain is from adding weather and calendar context to the PM2.5-only XGBoost model: MSE decreases by 3.764 and the confidence interval is fully below zero. Cosine retrieval improves over random retrieval in the retrieval-only setting. In contrast, adding cosine retrieval features to the already strong XGBoost context model does not improve the all-origin result in this run. Event-hybrid retrieval and drift indicators also do not show a clear overall improvement beyond the context baseline. The M12 drift gate improves

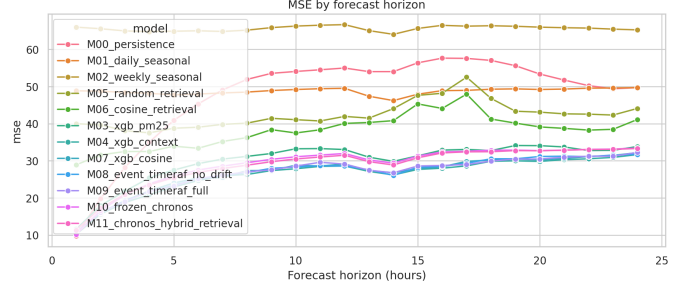


Fig. 5. MSE by forecast horizon on the test set.

MAE relative to M04 with a fully negative confidence interval (-0.0097 , 95% CI $[-0.0185, -0.0013]$), but its MSE interval crosses zero. The auxiliary A01 source is an XGBoost random-retrieval variant used only for the drift-gate selection analysis. The M12 result should therefore be described as the best numerical model rather than a statistically decisive MSE win.

The k sensitivity results in Table X show that the retrieval-only cosine baseline improves as more neighbors are used within the tested range. The best retrieval-only configuration among $k \in \{1, 4, 8, 16\}$ is $k = 16$, with MSE 36.488. The reported Event-TimeRAF feature models used the pre-specified default $k = 8$.

Table XI reports representative event and drift subsets for the strongest context model, the full Event-TimeRAF model, the drift-gated Event-TimeRAF variant, and the two Chronos variants. All subsets satisfy the pre-specified minimum sample-size criterion. The event subset has lower MSE than the non-event subset for these models, which should not be interpreted as events being easier in general; it reflects the particular 2024 test-period event hours retained by the audit. On drift-flagged origins, M12 matches the M04 MSE because the gate selects the context model for that regime, while M12 improves the non-drift and event subsets slightly through its non-drift source.

Fig. 5 and Fig. 6 show horizon-wise test errors. Fig. 7 summarizes retrieval behavior, and Fig. 8 shows the drift-score distribution across the test period. These plots are generated directly from a consistent set of saved artifacts identified by 20260723T112033170131Z.

TABLE VI
EXPERIMENTAL HYPERPARAMETERS AND RUNTIME SETTINGS

Component	Setting
Study period	2019–2024
Lookback and horizon	$L = 168$ hours, $H = 24$ hours
Split	70% train, 15% validation, 15% test, chronological
Retrieval	$k = 8$, sensitivity for $k \in \{1, 4, 8, 16\}$
Knowledge-base stride	192 hours
Hybrid retrieval weights	time-series 0.5, weather 0.2, calendar 0.1, event 0.2
XGBoost	350 estimators, max depth 6, learning rate 0.05, subsample 0.85, colsample 0.85, $\lambda = 1.0$
Drift detector	24-hour recent window, 168-hour reference window, 0.90 validation quantile threshold
Chronos checkpoint	amazon/chronos-bolt-small, batch size 64
Chronos fusion weights tested	$\{0, 0.25, 0.5, 0.75, 1.0\}$; selected 0.75
Bootstrap	500 paired block-bootstrap resamples, 24-hour blocks

TABLE VII
METHODOLOGICAL POSITIONING AGAINST RELATED FORECASTING WORK

Method family	TSFM	Retrieval	Events	Explainability	Comment
PatchTST-style Transformers [21]	No	No	No	Limited	Strong sequence modeling, but event context is not central.
Chronos [5]	Yes	No	No	Limited	Frozen zero-shot baseline used in this study.
TimeRAF [9]	Yes	Yes	No	Retrieval cases	Base retrieval-augmented TSFM method; does not target official event-aware PM2.5.
XGBoost context baseline [10]	No	No	No	Feature importance	Strong supervised tabular baseline for local PM2.5.
Event-TimeRAF	Yes	Yes	Yes	Evidence-grounded	Adds official event audit, temporal leakage control, drift indicators, and Chronos retrieval fusion.

TABLE VIII
OVERALL TEST RESULTS FOR 24-HOUR PM2.5 FORECASTING

Model	MSE	MAE	RMSE	R^2
M00 persistence	47.649	4.137	6.903	-0.130
M01 daily seasonal	48.706	4.057	6.979	-0.155
M02 weekly seasonal	65.606	5.123	8.100	-0.556
M03 XGBoost PM2.5	29.949	3.368	5.473	0.290
M04 XGBoost context	26.185	3.125	5.117	0.379
M05 random retrieval	42.367	4.257	6.509	-0.005
M06 cosine retrieval	38.083	3.940	6.171	0.097
M07 XGBoost cosine	26.672	3.151	5.164	0.367
M08 Event-TimeRAF no drift	26.677	3.145	5.165	0.367
M09 Event-TimeRAF full	26.712	3.149	5.168	0.367
M10 frozen Chronos-Bolt	28.941	3.205	5.380	0.314
M11 Chronos + retrieval	28.709	3.209	5.358	0.319
M12 drift-gated Event-TimeRAF	26.111	3.116	5.110	0.381

TABLE IX
SELECTED PAIRED BLOCK-BOOTSTRAP ABLATION RESULTS

Comparison	MSE diff.	95% CI low	95% CI high
M04 minus M03, weather/calendar	-3.764	-5.856	-2.071
M06 minus M05, cosine/random retrieval	-4.284	-5.862	-2.759
M07 minus M04, retrieval features	0.487	0.015	1.028
M08 minus M07, event hybrid	0.005	-0.292	0.275
M09 minus M08, drift indicators	0.035	-0.101	0.161
M09 minus M04, full Event-TimeRAF	0.528	-0.026	1.182
M11 minus M10, Chronos retrieval	-0.233	-0.665	0.152
M12 minus M04, drift gate	-0.073	-0.244	0.107
M12 minus A01, auxiliary gate source	-0.062	-0.182	0.031

A. Explainability Analysis

Explainability is implemented as a source-grounded evidence record rather than as a post-hoc free-text interpretation.

For each evaluated forecast origin, the pipeline stores the run identifier, window identifier, origin timestamp, retrieved historical evidence identifiers, event evidence identifiers, leading

TABLE X
COSINE RETRIEVAL SENSITIVITY TO k

k	MSE	MAE	RMSE
1	65.129	5.133	8.070
4	41.237	4.134	6.422
8	38.083	3.940	6.171
16	36.488	3.851	6.040

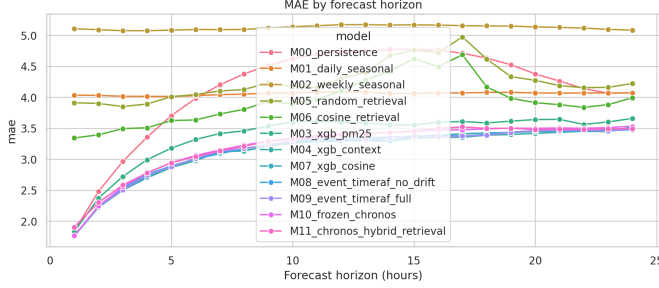


Fig. 6. MAE by forecast horizon on the test set.

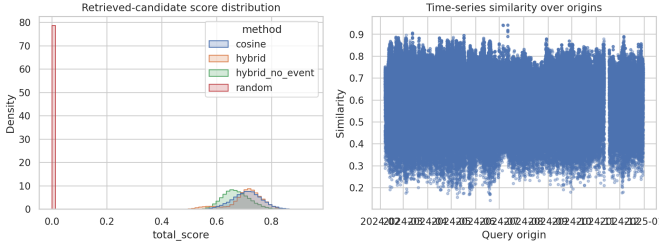


Fig. 7. Retrieval diagnostics on the test set.

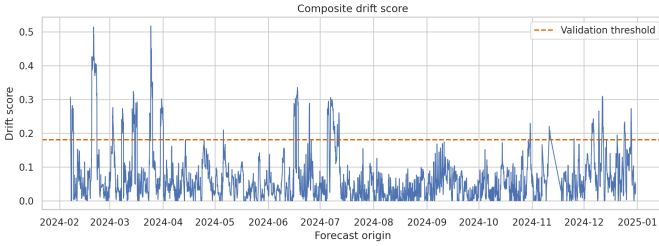


Fig. 8. Drift-score diagnostics on the test set.

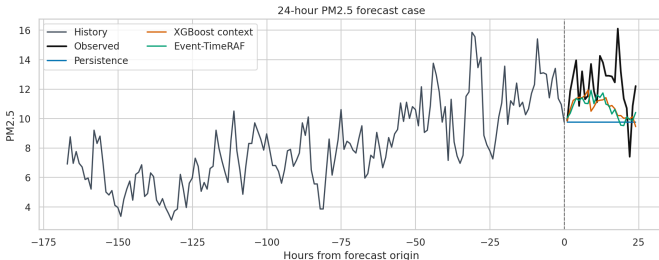


Fig. 9. Representative forecast case from the test set.

feature effects, drift components, retrieval spread, validation residual MAE, uncertainty proxy, drift score, drift flag, and event-availability mode. The evaluation produced 7,199 explanation records in `explanations.parquet`. A represen-

tative record identifies the strongest local contribution fields as the current PM2.5 value and recent rolling PM2.5 means, links the prediction to three retrieved historical windows, links recent source-recorded storm events when available, and marks distribution shift when the drift detector is active.

Table XII summarizes the explanation layer. This design supports auditability because every explanatory statement can be traced back to a model feature, a retrieved time-series window, an official event record, or a drift component. It does not prove causality: event evidence is reported as contextual support, and the drift flag is treated as an uncertainty warning.

B. Computational Cost and Reproducibility

The complete experiment executed in 1,576.9 seconds in a cloud-hosted Linux environment using Python 3.12.13. The recorded package versions include NumPy 2.0.2, pandas 2.3.3, scikit-learn 1.6.1, XGBoost 3.2.0, PyTorch 2.10.0+cu128, and `chronos-forecasting` 2.3.1. The run manifest records the configuration SHA-256 hash and artifact checksums. The CPU, GPU, and RAM model identifiers were not saved in the manifest; therefore this paper reports runtime and software versions but does not claim hardware-normalized efficiency.

The main interpretation is that conventional supervised context features are very strong for this Los Angeles County 24-hour PM2.5 task. Retrieval is useful as a standalone historical-pattern baseline, gives a small favorable MSE change when fused with frozen Chronos-Bolt, and improves the best numerical result when used through a drift-gated selector. However, the present feature-level Event-TimeRAF design does not beat the best XGBoost context baseline overall. Event-TimeRAF should therefore be presented as an implemented and audited event-aware retrieval framework with selective gains, not as a uniformly superior forecasting model.

Finally, all event-aware interpretations must be read with the availability caveat. NOAA Storm Events records are official and hash-verified, but the source does not provide machine-readable publication timestamps. The experiment records event start as the availability time, so event-related results are retrospective sensitivity results. A strict operational study would require event or smoke sources with genuine issue or publication times, such as an audited NOAA HMS cache or alert product archive.

VIII. LIMITATIONS AND FUTURE WORK

The most important limitation is event availability. NOAA Storm Events records are official and source-verified, but the public detail files do not provide machine-readable publication timestamps. The event-aware component therefore uses event start time as the availability proxy. This makes the reported event-aware results retrospective sensitivity results rather than strict operational nowcasts. A production deployment would require sources with auditable issue times, such as alert archives, smoke products, or an HMS cache with verified timestamps.

The second limitation is external validation. The experiment evaluates one carefully audited setting: Los Angeles County PM2.5 from 2019–2024. This is appropriate for an initial

TABLE XI
REPRESENTATIVE EVENT AND DRIFT SUBSET RESULTS

Subset	Model	Origins	MSE	MAE
Non-event	M04 XGBoost context	6,667	27.520	3.179
Event	M04 XGBoost context	532	9.451	2.450
Non-event	M09 Event-TimeRAF full	6,667	27.942	3.187
Event	M09 Event-TimeRAF full	532	11.301	2.666
Non-event	M12 drift-gated Event-TimeRAF	6,667	27.447	3.169
Event	M12 drift-gated Event-TimeRAF	532	9.375	2.450
Non-drift	M04 XGBoost context	6,639	26.107	3.088
Drift	M04 XGBoost context	560	27.110	3.562
Non-drift	M09 Event-TimeRAF full	6,639	26.330	3.088
Drift	M09 Event-TimeRAF full	560	31.244	3.872
Non-drift	M12 drift-gated Event-TimeRAF	6,639	26.027	3.078
Drift	M12 drift-gated Event-TimeRAF	560	27.110	3.562
Drift	M10 frozen Chronos-Bolt	560	28.333	3.476
Drift	M11 Chronos + retrieval	560	28.198	3.580

TABLE XII
EXPLANATION EVIDENCE STORED BY EVENT-TIMERAF

Evidence field	Interpretation role
Top feature effects	Local model drivers from the direct forecaster.
Retrieved windows	Historical analogues used by the retrieval module.
Event evidence IDs	Official storm-event context linked to the origin.
Drift components	Mean, variance, weather, similarity, and event-burst shift signals.
Uncertainty proxy	Combined retrieval spread and validation residual error.

TABLE XIII
COMPUTATIONAL AND REPRODUCIBILITY SUMMARY

Item	Recorded value
Run identifier	20260723T112033170131Z
Runtime	1,576.9 seconds
Platform	Cloud-hosted Linux environment
Python	3.12.13
Forecast horizon	24 hours
Lookback window	168 hours
Bootstrap resamples	500
Chronos checkpoint	amazon/chronos-bolt-small
Code availability	https://github.com/shasabbir/event-time-raf

site-specific evaluation, but it does not establish geographic generalization. Future work should repeat the same protocol on another U.S. county or city with different meteorology, emissions, and event profiles.

The third limitation concerns model integration. Event-TimeRAF currently uses retrieved trajectories and event context as feature-level evidence, a lightweight validation-selected drift gate, and a frozen Chronos-Bolt evaluation. This provides less expressive retrieval integration than a learnable retrieval-and-prompting architecture that directly conditions the foundation model. Future work should test a learnable retriever, richer event-type embeddings, calibrated uncertainty estimation, and a closer adaptation of TimeRAF-style Channel Prompting for environmental forecasting.

IX. CONCLUSION

This paper presented Event-TimeRAF, an event-aware retrieval-augmented framework for explainable Los Angeles County PM2.5 forecasting under distribution shift. The implemented pipeline combines official EPA PM2.5 observations, NOAA weather, hash-verified NOAA Storm Events records, leakage-safe time-series retrieval, direct XGBoost forecasters, drift indicators, evidence-based explanations, and a frozen Chronos-Bolt evaluation. In the 2019–2024 experiment, the drift-gated Event-TimeRAF variant obtains the best numerical all-origin result with MSE 26.111, MAE 3.116, RMSE 5.110, and R^2 0.381. The strongest ungated supervised context model obtains MSE 26.185, while the full feature-level Event-TimeRAF XGBoost variant obtains MSE 26.712. Retrieval improves the standalone cosine baseline over random retrieval and gives a small favorable MSE change when fused with frozen Chronos-Bolt. The M12 drift gate also improves MAE over the context baseline with a negative bootstrap interval, although its MSE improvement is not statistically conclusive. The study therefore positions Event-TimeRAF as an auditable event-aware retrieval framework with selective empirical gains. Stricter event availability, external validation, and deeper foundation-model integration remain priorities for future research.

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